When the silk worm is fully grown it starts spinning its cocoon on straw placed on the trays. The silk fluid from the special glands issues from two holes, one on either side of the head called **spinnerets**.

The fluid hardens as it comes in contact with the air and two long fibres which are stuck together with silk gum are formed. The cocoons are heated to kill the pupa inside, otherwise the moths would destroy some of the silk. Some are allowed to become moths to provide eggs.

i. Manufacturing process:

The manufacturing process involves Reeling, Throwing, Degumming, Weaving, Dyeing and sometimes weighting.

a) **Reeling:** This is the process of unwinding the silk filament from the cocoon. The cocoons are boiled in water to soften the gum so as to unwind the filaments.

b) **Throwing:** Throwing is the process of combining several reeled strands to make a yarn. The number of strands are twisted together to form a strong yarn.

c) **Degumming:** The gum left on the fibres to protect them are now removed by boiling in soap and water. Sometimes degumming is left until the fabric is woven. Weaving is carried out the same way as for other fibres.

d) **Weighting:** Weighting is the process of treating silk with certain metallic salts to give weight and body to the product.

ii. Properties:

Shape	:	Like glass rod
Length	:	30000 to 39000 cm
Elasticity	:	Good
Stretch ability	:	Good
Flexibility	:	Medium
Density	:	1.25g/cm^3
Reaction	:	Causes less damage with acid.
To Flame	:	Burns with a sputtering flame.
Sunlight	:	Increased exposures will break down the fibre.
Insects	:	Destroyed by Carpet beetles.
Drying	:	Quick and good.
Dyeing	:	good absorption for acid and basic dyes.

iii. Uses:

- Silk fabrics are noted for their soft luxurious handle, rich lustre, warmth, resilience and crease resistance, strength and excellent draping quality.
- A wide range of fabrics are made ranging from sheer chiffon to firmer dress and suiting material to heavy brocades to the rich pile velvet.
- Silk serves best for ceremonial occasions, evening or day ware and lingerie.

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